ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the official syllabus sentence (quadratics with real coefficients), power-sum recurrences and closed forms for $\alpha^n+\beta^n$ are extension theory. Nothing here is needed for Part I; the habit — never solve for roots when a symmetric combination is asked — is the transferable skill.

Power sums from coefficients alone

SHORT DERIVATION

Newton's recurrence for two roots

α,β satisfy $ax^2+bx+c=0$. Multiply the equation for α by α^{n-2}, the one for β by β^{n-2}, and add. With $S_n=\alpha^n+\beta^n$:

$$aS_n+bS_{n-1}+cS_{n-2}=0, \qquad S_0=2, \quad S_1=-\tfrac{b}{a}$$

So $S_n=\frac{-bS_{n-1}-cS_{n-2}}{a}$ generates every power sum from the coefficients — no radicals, ever. Start at $S_0=2$, not S_1 (checklist item).

WORKED EXAMPLE

Worked — Newton's sums

$x^2-5x+3=0$: $S_0=2$, $S_1=5$;
$S_2=5\cdot5-3\cdot2=19$;
$S_3=5\cdot19-3\cdot5=80$. Matches $S^2-2P=19$ and $S^3-3PS=125-45=80$ ✓ (and numerically $\alpha\approx4.3028,\beta\approx0.6972$ give $\alpha^3+\beta^3\approx80.00$ ✓).

KEY RESULT

The golden quadratic generates Fibonacci and Lucas

For $x^2-x-1=0$ with roots $\varphi=\frac{1+\sqrt5}{2}$, $\psi=\frac{1-\sqrt5}{2}$: induction on $\alpha^2=\alpha+1$ gives

$$\alpha^n=F_n\alpha+F_{n-1} \qquad (F_0=0,F_1=1)$$

for both roots. Adding the two identities: $S_n=\varphi^n+\psi^n=F_n+2F_{n-1}=F_{n+1}+F_{n-1}=$, the **Lucas numbers** $2,1,3,4,7,11,18,29,47,\ldots$ Subtracting and dividing by $\sqrt5=\varphi-\psi$ gives **Binet**: $F_n=\frac{\varphi^n-\psi^n}{\sqrt5}$. Worked check: $F_{10}=55$, and $55\varphi+34\approx122.9918\approx\varphi^{10}$ ✓.

STANDARD APPLICATION

Standard applications

Integer-valuedness: if S,P are integers, every S_n is an integer (recurrence), so $\varphi^n+\psi^n\in\mathbb{Z}$ despite the radicals — the standard trick for proving $\lfloor\varphi^{2n}\rfloor$-type results, using $|\psi|<1$ so $\psi^n\to0$ and the integer L_n sits within 1 of φ^n.

COMMON TRAP

Common traps — power sums

- Recurrence sign: $aS_n+bS_{n-1}+cS_{n-2}=0$ — move terms carefully.
- $S_0=2$ (two roots), including when a root is repeated.
- Binet needs $\varphi-\psi=\sqrt5$ in the denominator, not $2\sqrt5$.

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

The discriminant as $a^2(\alpha-\beta)^2$, rational-root criteria, and the resolvent solution of the cubic are classical extension material beyond the official quadratic-equations syllabus line.

KEY RESULT

What the discriminant actually measures

$$\Delta = b^2 - 4ac = a^2(\alpha - \beta)^2$$

The factor $a^2 > 0$ is pure scale: the **sign** of Δ is the sign of $(\alpha - \beta)^2$ — positive (distinct real), zero (coincident), negative (non-real conjugates, where $\alpha - \beta$ is purely imaginary). The **value** measures root separation. One number, complete information.

KEY RESULT

Rational roots are a perfect-square question

For rational a, b, c: roots rational $\Leftrightarrow$ Δ is the square of a rational. Worked: $x^2 - kx + (k-1) = 0$ has $\Delta = k^2 - 4k + 4 = (k-2)^2$, a perfect square for **every** integer k; roots $x = \frac{k \pm |k-2|}{2} = 1, k-1$ (indeed $x^2 - kx + k - 1 = (x-1)(x-k+1)$ ✓).

SHORT DERIVATION

Lagrange's resolvent — a cubic solved by one quadratic

For $x^3 + px^2 + qx + r = 0$, the shift $x = y - \frac{p}{3}$ kills the square term: depressed cubic $y^3 + Py + Q = 0$. Set $y = u + v$; then $y^3 = u^3 + v^3 + 3uvy$, so matching needs $3uv = -P$ and $u^3 + v^3 = -Q$. Hence $u^3 v^3 = -\frac{P^3}{27}$, and u^3, v^3 are the roots of the **resolvent quadratic**

$$t^2 + Qt - \frac{P^3}{27} = 0$$

Solve it, take cube roots chosen with $uv = -\frac{P}{3}$ (the pairing constraint is essential — arbitrary cube roots give wrong y), and $y = u + v$. Worked: $x^3 - 6x^2 + 11x - 6 = 0$: $x = y + 2$ gives $y^3 - y = 0$; resolvent $t^2 + \frac{1}{27} = 0$; the three admissible (u, v) pairings give $y = 1, 0, -1$, so $x = 1, 2, 3$ ✓ (sum $6 = -p$, product $6 = -r$ ✓).

INTUITION

Why the ladder stops at degree 4

Quadratic → cubic (resolvent quadratic) → quartic (resolvent cubic): each degree is solved by the previous. At degree 5 the ladder provably ends — the general quintic is not solvable by radicals (Abel–Ruffini), and the explanation of why is Galois theory. The resolvent is the last rung reachable by pure algebra.

COMMON TRAP

Common traps — resolvents and discriminants

- Taking cube roots of u^3, v^3 independently: the three choices must satisfy $uv = -\frac{P}{3}$.
- "Rational coefficients" hypothesis dropped: Δ a perfect square decides rationality only over $\mathbb{Q}$.
- $\Delta = a^2(\alpha - \beta)^2$, not $(\alpha - \beta)^2$ — the a^2 matters for exact values.

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IMO-level)

Vieta jumping is a number-theoretic descent technique, far beyond the JEE syllabus. It is included because its engine is exactly Part I material: Vieta's relations plus root-flipping.

KEY RESULT

The move, isolated

Given a symmetric condition on positive integers (a, b) with a fixed integer quotient k: read the condition as a **quadratic in one variable**. One root is a; Vieta hands you the other root $a' = (\text{sum of roots}) - a = \frac{\text{product}}{a}$, automatically an integer with the same k. If $0 \le a' < b \le a$, the pair (b, a') is a smaller solution with the same k — descend until forced to stop; the terminal configuration reveals what k must be.

SHORT DERIVATION

IMO 1988 Problem 6 — the famous application

Claim: if a, b are positive integers with $ab+1 \mid a^2+b^2$, then $k = \frac{a^2+b^2}{ab+1}$ is a perfect square. Fix such a k and view $a^2 - kba + (b^2 - k) = 0$ as a quadratic in a. The other root is $a' = kb - a = \frac{b^2-k}{a}$; an integer since $a \mid b^2 - k$ (rearrange the equation: $b^2 - k = a(kb - a)$). Ordering $A \geq B$, one checks $0 \leq a' < B$, so descent strictly shrinks the pair while k never changes. It can only halt at $a' = 0$, where the equation gives $k = B^2$ — a perfect square, and the same k as the original pair. ∎

WORKED EXAMPLE

Worked — tracing the descent

$(a, b) = (8, 30)$: $k = \frac{64+900}{241} = 4$. Order $A = 30 \geq B = 8$; other root $a' = 4 \cdot 8 - 30 = 2$, pair $(8, 2)$ (check $\frac{64+4}{17} = 4$ ✓). Reorder $A = 8, B = 2$; $a' = 4 \cdot 2 - 8 = 0$; halt with $k = 2^2 = 4$ ✓. Brute force over $1 \leq a, b < 300$: every admissible quotient is a perfect square ($1, 4, 9, 16, 25, 36, \ldots$) — no exceptions.

STANDARD APPLICATION

The $k = 9$ chain (building solutions up)

Descent runs from any solution down to $(0, 3)$ for $k = 9$; reversing $(a, b) \mapsto (b, kb - a)$ builds all solutions upward: $(0, 3) \to (3, 27) \to (27, 240) \to (240, 2133) \to$ Checks: $\frac{9+729}{82} = 9$ ✓; $\frac{729+57600}{6481} = 9$ ✓. So with $a, b \leq 500$: $(3, 27), (27, 3), (27, 240), (240, 27)$.

COMMON TRAP

Common traps — Vieta jumping

- Forgetting k is *invariant* under the flip — that invariance is the entire proof.
- Assuming a' is a non-negative integer without proof: integrality comes from Vieta (product/sum of roots), non-negativity from $a' = \frac{B^2-k}{A}$ plus the ordering argument; both need care in a written solution.
- Descending without a well-ordering: the descent terminates because positive integers cannot decrease forever — state it.